import os import sys # Script to build Teal theme HTML and Canva-editable PPTX poster for SignBridge html_content = """

SignBridge

A Real-Time Bidirectional Sign Language Communication System with a Unique Hand Gesture Dataset

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Different Hands
Same Language
A Brighter Future

1. MOTIVATION

- Sign language is a powerful and essential means of communication, breaking down barriers for individuals who are speech & hearing-impaired.
- The primary objective is to address the need for robust Indian Sign Language (ISL) recognition in diverse environments, including self-created datasets and benchmark data.
- Leverages CNN and LSTM to enhance recognition performance with higher accuracy, robustness and faster training.
- Provides an alternative to traditional ML and standalone CNNs by combining spatial and temporal feature learning.

2. HOW SIGNBRIDGE WORKS (WORKFLOW)

User (Shows gesture) → **Camera** (Captures video/ image stream) → **Preprocessing** (Hand isolation, resize, normalize) → **CNN + LSTM** (Feature extraction + sequence modeling) → **Classification** (26 classes) → **Output** (Text/Audio communication)

SAMPLE HAND GESTURES (Emoji Representation)

Dataset I – Self Created (Simple)

Dataset II – Studio (Uniform)

Dataset III – Complex (Non-uniform)

3. DATASET OVERVIEW & COMPARISON

Dataset	Background	Environment	Purpose	Classes	Images
Dataset I	Simple	Controlled	Self-created ISL	26	5,200
Dataset II	Simple studio	Uniform lighting & mask	Benchmark / comparison	26	-
Dataset III	Complex	Low light / non-uniform	Robustness evaluation	26	-

4. PREPROCESSING (HAND ISOLATION)



5. MODEL ARCHITECTURE (CNN + LSTM)



6. WHY CNN + LSTM?

CNN – Spatial Feature Extraction

- Learns hand shape, edges, visual features
- Converts image to 512-D feature vector
- Works well with variations in background & lighting

+

LSTM – Temporal Sequence Modeling

- Captures frame-to-frame changes
- Learns gesture sequence
- Handles long-range dependencies

= Better Accuracy + Robustness + Real-time Performance

7. MODEL PARAMETERS

224 × 224 Input Size	**512 / 768** Embedding Dim	**128** LSTM Units
0.3 Dropout	**32** Batch Size	**10 / 20 / 50** Epochs

8. KEY CONTRIBUTIONS

01 Self-created ISL gesture dataset (26 classes, 5,200 images)

02 Evaluation across simple and non-uniform environments

03 CNN + LSTM spatial-temporal modelling

04 High accuracy up to 100% (Dataset II)

9. PERFORMANCE ANALYSIS

(a) Accuracy Comparison Across Datasets



(b) State-of-the-Art Comparison

Dataset	Methodology	Accuracy (Existing)	Accuracy (Proposed)
Dataset I	HU Moments + GLCM + Random Forest	98.0%	**98.5%**
Dataset II	Deep Vision Transformers	99.29%	**100%**
Dataset III	CNN	99.56%	**99.8%**

10. REAL-WORLD ENVIRONMENTS

Simple Background
(Studio + Uniform Lighting)

Complex Background
(Low Light / Non-uniform)

Testing across controlled and non-uniform environments

11. SCOPE & SOCIETAL IMPACT

Scope

- ✓ Real-time sign language recognition
- ✓ Supports both static & dynamic gestures
- ✓ Can be extended to multiple languages
- ✓ Potential for mobile and wearable devices

Societal Impact

- ✓ Empowers speech & hearing-impaired community
- ✓ Improves accessibility and inclusion
- ✓ Bridges communication gaps
- ✓ Supports education & workplace accessibility

12. LIMITATIONS & FUTURE WORK

→ Limitations

- Limited dataset size
- Only 26 classes (A–Z)
- Focus on static gestures
- Performance may vary in extreme real-world conditions

→ Future Work

- Increase dataset size and diversity
- Include dynamic gestures & sentences
- Improve real-time deployment
- Extend to multi-language recognition

13. CONCLUSION & REFERENCES

13. CONCLUSION

The proposed **CNN-LSTM** architecture for static ISL recognition is a promising solution for the speech and hearing-impaired community. It achieves high accuracy, works in diverse environments, and ensures efficient training with fewer epochs.

14. REFERENCES

1. A. O. Hashi et al., "A systematic review of hand gesture recognition," *IEEE Access*, vol. 12, pp. 143603–143623, 2024.
2. U. Rastogi et al., "Advancements in Machine Learning for Gesture Recognition," *Journal of Computer Vision and AI*, 2023.

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"SignBridge – Connecting People Beyond Words"

""" html_path = r"c:\Users\Admin\Desktop\Poster\SignBridge_Poster_24×36_Teal_Canva_Editable.html" with open(html_path, "w", encoding="utf-8") as f: f.write(html_content) print("HTML file written successfully to:", html_path)